THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which statement about a ceiling fan switched on is correct? 1

(a) the blades move randomly **(b)** the fan is at rest
(c) each blade moves linearly **(d)** each blade moves in a circular path

2. Parallax error is least when the scale is: 1

(a) placed far away
(b) viewed from the side
(c) placed touching the object and viewed straight
(d) tilted

3. The thickness of 300 sheets of paper is 1.5 cm. The thickness of one sheet in mm is: 1

(a) 0.5 mm **(b)** 5 mm **(c)** 0.05 mm **(d)** 0.005 mm

4. A swing is described as: 1

(a) to and fro motion repeated at equal intervals
(b) linear motion only
(c) random motion
(d) rest

5. If 1 m is divided into 1000 equal parts, each part is a: 1

(a) kilometre **(b)** millimetre **(c)** decimetre **(d)** centimetre

6. Which is the best estimate of the height of a door? 1

(a) 2 cm **(b)** 2 mm **(c)** 2 km **(d)** 2 m

7. Which of these is not a periodic motion? 1

(a) clock hand **(b)** a falling coin **(c)** heart beat **(d)** a pendulum

8. The distance 0.25 km in cm is: 1

(a) 25000 cm **(b)** 25 cm **(c)** 250 cm **(d)** 2500 cm

9. **Assertion (A):** The motion of a swing is periodic. 1

Reason (R): A swing repeats its motion after equal intervals of time.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** To measure the length of a curved line, a thread can be used. 1
Reason (R): A thread is a rigid straight scale.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A pendulum has a time period of 1.5 s. How many oscillations occur in 3 minutes? 2

12. How many millimetres are there in 2.4 m? 2

13. Why is it better to measure 50 oscillations instead of 1 for a pendulum? 2

14. A student measured 5.4 cm, 5.5 cm, and 5.6 cm. What is the mean? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The Earth takes 24 hours for one rotation. Find the number of rotations in a week and in 30 days. 3

16. Why is it better to use a thread than a metre rod for measuring a curved path? 3

17. Which motion of a clock's parts is periodic? Give reasons. 3

OR

A pendulum swings through 40 oscillations in 1 min 20 s. Find the time period. 3

18. A car's odometer shows 12 345.6 km at start and 12 395.1 km after a journey. Find the distance travelled. 3

OR

Explain the use of a measuring wheel. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Karan wants to find the length of a curved road shown on a map and the circumference of a round plate. He has a ruler and a piece of thread. He puts the thread along the curve, marks the ends and then straightens the thread against the ruler. The thread measures 24 cm. 5
(i) Which tool is used to measure straight short lengths?
(ii) How many millimetres are in 24 cm?
(iii) Name the unit used for large road distances.
(iv) Why did Karan use a thread instead of a ruler directly? Why must the thread not be stretched?

20. Design an experiment to measure the circumference and diameter of five circular objects and discuss the pattern. 5

OR

Analyse the sources of error while measuring the length of a table with a 30 cm ruler. Suggest improvements. 5

END OF QUESTION PAPER — Total: 40 Marks